Xuanzhi Chen

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Research Interests

Spatial multi-omics, machine learning for biology, and scientific agents.

Education

Peking University Beijing, China
B.Eng. Candidate in Bioinformatics, School of Life Sciences *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA: 3.77/4.0.** Pursuing a double degree in Economics at the National School of Development.
- **Selected coursework:** Computational Immuno-oncology (99/100); Data Structures and Algorithms (98/100); Statistical Analysis of Genomics Data (A); AI Methods and Applications in Life Sciences (A).

Preprint

Nianping Liu, **Xuanzhi Chen**, Miao Cui, Xiaoying Liao, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. **Bio-Babel: autonomous cross-language reconstruction of agent-ready computational biology ecosystems.** *bioRxiv*, 2026. https://doi.org/10.64898/2026.08.30.743167.

Research Experience

College of Future Technology, Peking University Beijing, China
Undergraduate Researcher; Advisor: Prof. Hu Zeng *Jul. 2024 – Present*

- **STATES-GBM:** Built and optimized upstream processing for STATES, which jointly profiles spatial transcriptomes and translatomes. Benchmarked cell-segmentation methods, integrated 3D Cellpose nuclear segmentation into ClusterMap, and improved image stitching. Reduced seams misaligned by > 3 pixels from 14.2% to 0.3% and multi-nucleus segments from 28.8% to 13–14% against the same nuclear reference. Analyzed malignant-cell identities, subtypes, and spatial variation in translation efficiency in glioblastoma.
- **iST Agent:** Built an agent for reliable execution of heterogeneous imaging-based spatial transcriptomics workflows, separating source-derived contracts from memory of runtime failures and fixes. Generated contracts for seven pipelines and reproduced one pipeline's decoding byte for byte. In one comparison, preflight checks caught a multi-sample configuration defect with memory but missed it without memory.

Department of Genetics, Stanford University Stanford, CA
Undergraduate Researcher; Advisor: Prof. Xiaojie Qiu *Jun. 2026 – Present*

On-site summer research, Jun.–Aug. 2026; continued remotely.

- **Bio-Babel:** Built an annotation agent that generates machine-readable usage contracts for reconstructed packages, enabling downstream agents to access their functionality. Extended coding-agent compatibility and developed tooling for agent-driven Python-to-C++ reconstruction. Conducted the pancreatic differentiation case study, including preprocessing and cell annotation, followed by analysis through Bio-Babel's agents. Reviewed trajectory analyses and completed genotype-aware screening and biological interpretation of reduced maturation signatures under SWI/SNF loss.
  https://github.com/Bio-Babel/UMI-tools-cpp | https://github.com/Bio-Babel/bio-babel-pancreas
- **End-to-end spatial transcriptomics modeling:** Developing and training a model on public MERFISH data that couples molecular localization and barcode decoding with cell segmentation and molecule-to-cell assignment. The architecture combines RNA fluorescence and cellular morphology, learning cross-round correspondences without a separate image-warping stage. The project tests whether sharing evidence across these interdependent tasks can reduce errors introduced by sequential processing.

Canchen Technology Remote
Research Intern; Mentor: Dr. Lei Li *Oct. 2026 – Present*

- **Agentic virtual tissue modeling:** Investigating hybrid world models that integrate foundation models with mechanistic simulations, connecting learned cell-state representations with explicit biological dynamics.
- **Agent-guided prediction:** Designing bidirectional feedback between foundation models and LLM agents, with trainable interfaces for agents to modulate features, select model components, and iteratively refine predictions using literature and biological context.

Selected Presentations

Building an Agentic Tissue World Model.
Journal club presentation, Canchen Technology, Nov. 2026.

Bio-Babel.
Invited oral presentation, PKU Undergraduate Honors Program in Biology Annual Symposium, Oct. 2026.

Bio-Babel and End-to-End Spatial Transcriptomics Modeling.
Research presentation, Qiu Lab, Stanford University, Aug. 21, 2026.

How to Build the Virtual Cell?
Group course presentation, Systems Biology, Peking University, Jun. 6, 2026.

SpatialFusion: A Lightweight Multimodal Foundation Model for Pathway-Informed Spatial Niche Mapping.
Journal club presentation, Zeng Lab, Peking University, May 27, 2026.

Honors and Awards

Academic Excellence Award, Peking University	2026
Peking University Scholarship	2025
Peking University Freshmen Scholarship	2023
Gold Medal, China National Biology Olympiad Final	2022

Technical Skills

Experimental: In situ spatial omics (STATES), molecular cloning, CRISPR-Cas9, PCR/qPCR, Western blotting.

Computational: Python, R; single-cell and spatial omics, image analysis, scientific agents, workflow automation.